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Inventor(s) Yassin; Mohammad Saleh et al.

SYSTEMS, METHODS, AND COMPUTER PROGRAM PRODUCTS FOR PROVIDING IMMUTABLE DIGITAL TESTIMONY

Abstract

Methods, systems and computer program products are provided for creating and retrieving an immutable digital testimony involving activating a process on a device with a unique identifier and authenticating the creator of an immutable digital testimony using a unique user ID registered with a testimony network. The device activates associated cameras to generate a media stream. Frames are hashed using a selected scheme to create stream hashes, forming a live stream hash, and capturing metadata to be recorded on a blockchain. The stream media and hashing scheme are transmitted to a cloud server for storage, with cloud or distributed file system (DFS) addresses retrieved to determine where the media is saved. The live stream media is encrypted and sent to the DFS, encrypted media addresses are received, and the stream media is saved locally on the device.

Inventors: **Yassin; Mohammad Saleh (Miami, FL), Fernandez Chiques; Carlos Damian (Miami, FL)**

Applicant: **ALIBI LLC** (Miami, FL)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and is a Continuation of U.S. patent application Ser. No. 18/733,533, filed Jun. 4, 2024, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/506,185, filed Jun. 5, 2023, which are hereby incorporated by reference in their entirety herein.

TECHNICAL FIELD
[0002] Example embodiments described herein relate generally to signal processing and blockchain technologies, and more particularly to performing signal processing on various signals to provide immutable digital testimony.
BACKGROUND
[0003] There are many situations for which it may be necessary to provide evidence that demonstrates a person is at a place they say they are or were. An alibi, for example, is typically defined as a claim or piece of evidence that one was elsewhere when an act, such as a criminal one, is alleged to have taken place. An alibi defense can have witnesses testify and present evidence at trial to support an alibi defense. Certain industry, government and personal situations may require such evidence outside the criminal context as well.
[0004] Evidence of someone's location or alleged conduct can be difficult to prove, however. One person's testimony is often not enough. If the visual evidence and location cannot be proven with a relatively high degree of certainty, then it may not convince those requiring proof of the person's whereabouts and conduct at a particular time (e.g., parent, spouse, employer, government, court, etc.).
[0005] Just because a person declaring their whereabouts has evidence to support their claim does not automatically mean that the evidence they present will be accepted. One test evidence must meet, in order to be deemed admissible, is that the evidence must be authentic.
[0006] Authentication generally refers to a rule of evidence which requires that evidence must be sufficient to support a finding that the matter in question is what its proponent claims. The “authenticity” rule relates to whether the subject of an evidentiary offering (generally a tangible thing), is what it purports to be. This is a legal way of saying that evidence must be proven to be genuine to be admissible.
[0007] Authentication of evidence may be accomplished by technical means such as by presenting logs of mobile device communications obtained from a wireless network provider or by the presenting photographs or videos containing geolocation metadata. However, concerns about the authenticity of such evidence are of great concern. Technology has enabled even children with the ability to alter audio-visual content and related metadata contained in digital files. More concerning, recent advances in artificial intelligence are expected to usher in easier and automated means of altering or computer-generating digital videos and photos.
[0008] Typical technologies that attempt to thwart tampering of digital evidence attempt to detect clear indicators of alteration, such as by detecting unusual artifacts and inconsistencies in a digital file, or by embedding “invisible” data (e.g., watermarks, noise, and the like) into the digital file (e.g., an image or audio file). A playback device can then detect alterations of the content in the digital file or embedded data to provide an indication that the evidence has been altered.
[0009] However, typical techniques still have one concept in common. They modify the content and hence alter the evidence. Other techniques may not modify the content, but also cannot

authenticate that the content has not been modified and hence cannot authenticate the evidence with confidence. Accordingly, one technical problem involves using available technologies, such image capture devices, microphones, GPS, or other types of sensors in a manner to authenticate the content being used as evidence without altering the content. Another technical challenge involves creating a system that enables orchestration of a process for capturing, saving, and communicating over one or more networks, immutable evidence practically and efficiently.

[0010] At its core, a blockchain is a distributed ledger that records transactions. A public blockchain typically refers to blockchain that is a distributed public ledger, whereas a private blockchain, which may also be referred to as a managed blockchain, is a permissioned blockchain controlled by a single organization.

[0011] Blockchain technologies have been used to prepare media content for authentication by adding a block in a blockchain for a media content file. A block identifier from the blockchain network is generated and used as a watermark payload that is embedded into the content file. Thus, similar to the examples provided above, the evidence is modified.

[0012] NFTs, or Non-Fungible Tokens, are cryptographic assets held on a blockchain. Typically, an NFT is a unique digital certificate that functions as proof of ownership, or authenticity, of a digital artwork or asset. It is not the artwork or asset itself. Neither the NFT nor the underlying artwork, however, can be used as immutable evidence of one's location at a particular time.

SUMMARY

[0013] The example embodiments described herein meet the above-identified needs by providing methods, systems and computer program products for providing immutable digital testimony.

[0014] In an embodiment, a method for creating an immutable digital testimony is provided. The method involves the steps of: activating an immutable digital testimony creation operation on a device having an immutable digital testimony device identifier (ID), wherein the immutable digital testimony device ID is a unique identifier of the device; authenticating a creator of an immutable digital testimony based on an immutable digital testimony creator ID, wherein the immutable digital testimony creator ID is a unique user identification created upon registering with an immutable digital testimony network; capturing the immutable digital testimony device ID of the device used to create the immutable digital testimony; and upon authentication of the creator, initiating by the device, an immutable digital testimony capture operation including: activating one or more cameras associated with the device and causing the device to generate immutable digital testimony stream media, hashing one or more frames of the immutable digital testimony stream media for the one or more cameras by applying a selected immutable digital testimony hashing scheme from a plurality of hashing schemes, thereby creating one or more immutable digital testimony stream hashes, forming a live stream hash from the one or more immutable digital testimony stream hashes, capturing immutable digital testimony metadata to be recorded on an immutable digital testimony authentication blockchain, transmitting the immutable digital testimony stream media and the selected immutable digital testimony hashing scheme by the device to a cloud server, causing the selected immutable digital testimony hashing scheme and the immutable digital testimony stream media to be stored on the cloud server as immutable digital testimony cloud media, retrieving one or more immutable digital testimony addresses, each of the one or more immutable digital testimony addresses being one or more cloud addresses on the cloud server, one or more distributed file system DFS addresses on the DFS, or a combination of i and ii, correspondingly, wherein each of the one or more immutable digital testimony addresses corresponds to a location at which the immutable digital testimony cloud media is saved, the immutable digital testimony cloud media being at least a portion of the immutable digital testimony stream media, upon completion of the transmitting: encrypting the immutable digital testimony live stream media thereby creating encrypted immutable digital testimony live stream media, the encrypted immutable digital testimony live stream media being at least a portion of the immutable digital testimony stream media, sending to the DFS the encrypted immutable digital testimony live stream media to save the encrypted immutable digital testimony live stream media, receiving one or more immutable digital testimony DFS addresses, wherein each of the one or more immutable digital testimony DFS addresses corresponds to a location at which the encrypted immutable digital testimony live stream media has been saved, and saving the immutable digital testimony stream media on the device to create immutable digital testimony local media.

[0015] In some embodiments, the method involves: selecting one or more frames of the immutable digital testimony local media corresponding to the same one or more frames of the immutable digital testimony stream media, hashing one or more frames of the selected one or more frames of the immutable digital testimony local media using the selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony local hashes, forming a local symbiotic hash from the one or more immutable digital testimony local hashes, sending, by the device to a smart contract, the local symbiotic hash for validating against the live stream hash, and sending, by the device to the smart contract, the live stream hash, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, and the one or more immutable digital testimony cloud addresses.

[0016] In some embodiments, the method involves: selecting one or more frames of the immutable digital testimony cloud media corresponding to the same one or more frames of the immutable digital testimony stream media, hashing one or more frames of the selected one or more frames of the immutable digital testimony cloud media using the selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony cloud hashes, forming a cloud media hash from the one or more immutable digital testimony cloud hashes, and sending, by the cloud server to the smart contract, the cloud media hash for validating against the live stream hash.

[0017] In some embodiments, the method further involves creating, by the smart contract, a creator token by combining the immutable digital testimony creator ID and the immutable digital testimony device ID; creating, by the smart contract, an immutable digital testimony identifier (ID) by hashing the creator token with the live stream hash; and comparing, by the smart contract, the live stream hash, the local symbiotic hash, and the cloud media hash, and if a) the live stream hash mirrors the local symbiotic hash and b) the live stream hash mirrors the cloud media hash: creating, by the smart contract, an immutable digital testimony hash by hashing the immutable digital testimony ID, the one or more immutable digital testimony cloud addresses, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, and the live stream hash, recording, by the smart contract: the immutable digital testimony ID, the one or more immutable digital testimony cloud addresses, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, the immutable digital testimony hash, the live stream hash on the immutable digital testimony authentication blockchain, and the user permissions on an immutable digital testimony authentication blockchain, sending the user permissions to an immutable digital testimony orchestrator to create, based on the user permissions, a log entry indicating whether a corresponding immutable digital testimony creator ID has any rights over the immutable digital testimony cloud media that is associated with a specific immutable digital testimony ID, and recording, by the smart contract, the immutable digital testimony ID and immutable digital testimony hash on a public blockchain.

[0018] In some embodiments, the method further involves granting, by the smart contract to a requestor, viewing permission to an immutable digital testimony corresponding to the immutable digital testimony ID and registering the viewing permission on an immutable digital testimony authentication blockchain and the immutable digital testimony orchestrator.

[0019] In some embodiments, the method involves periodically validating and authenticating the immutable digital testimony authentication blockchain, using the immutable digital testimony hash recorded on the public blockchain.

[0020] In some embodiments, the method involves: if a) the live stream hash and the local symbiotic hash do not match or b) the live stream hash and a cloud media hash do not match: creating an alternate immutable digital testimony by saving the immutable digital testimony local media to the DFS and the cloud server and retrieving the one or more alternate immutable digital testimony DFS addresses and the one or more alternate immutable digital testimony cloud addresses, correspondingly, and providing a notification, by the smart contract, that the immutable digital testimony stream media was corrupted during the transmission or storage process.

[0021] In some embodiments, the method involves: sending, by the device, the immutable digital testimony metadata, the live stream hash, the local symbiotic hash, the one or more alternate immutable digital testimony cloud addresses, and the one or more alternate immutable digital testimony DFS addresses, to the smart contract.

[0022] In some embodiments, the method involves: receiving, by the smart contract, the alternate immutable digital testimony metadata, the live stream hash, the local symbiotic hash, and the one or more alternate immutable digital testimony DFS addresses; creating, by the smart contract, an alternate creator token by combining the immutable digital testimony creator ID and the immutable digital testimony device ID; creating, by the smart contract, an immutable digital testimony alternate identifier (ID) by hashing the alternate creator token with the live stream hash and adding ALT at the end of the immutable digital testimony alternate ID; creating, by the smart contract, the alternate immutable digital testimony hash by hashing the immutable digital testimony alternate ID, the immutable digital testimony metadata, the one or more alternate immutable digital testimony cloud addresses, the one or more alternate immutable digital testimony DFS addresses, the local symbiotic hash, and the live stream hash; recording, by the smart contract, the immutable digital testimony alternate ID, the immutable digital testimony metadata, the one or more alternate immutable digital testimony cloud addresses, the one or more alternate immutable digital testimony DFS addresses, the live stream hash, the local symbiotic hash, and the alternate immutable digital testimony hash to the immutable digital testimony authentication blockchain; recording, by the smart contract, user permissions corresponding to the immutable digital testimony alternate ID on the authentication blockchain and sending the user permissions to an immutable digital testimony orchestrator to update log entries of the immutable digital testimony orchestrator; and recording, by the smart contract, the alternate immutable digital testimony ID and alternate immutable digital testimony hash on the public blockchain.

[0023] In some embodiments, the method involves: receiving, by the immutable digital testimony orchestrator, user viewing permissions to an immutable digital testimony corresponding to the immutable digital testimony alternate ID; and creating, by the immutable digital testimony orchestrator, a local ledger of permission details based on the user permissions.

[0024] In some embodiments, the method involves: authenticating a third-party immutable digital testimony requestor; initiating, by the third-party immutable digital testimony requestor from an immutable digital testimony creator, a third-party request, the third-party request being any one or more of: an immutable digital testimony request, requesting to create and share a new immutable digital testimony, an immutable digital testimony view request, requesting permission to view an existing immutable digital testimony of the creator, or an immutable digital testimony share request, requesting permission to share an existing immutable digital testimony of the creator with another immutable digital testimony network user; creating, by the device of the third-party immutable digital testimony requestor, the third-party request by combining any one or a combination of: the third-party immutable digital testimony requestor ID, a phone number of an immutable digital creator, and an immutable digital testimony ID, and sending, by the third-party immutable digital testimony requestor, the third-party request to an immutable digital testimony orchestrator.

[0025] In some embodiments, the method further involves creating an immutable digital testimony trail that identifies immutable digital testimony created along a path, by: creating, by an immutable digital testimony orchestrator, a map of an immutable digital testimony trail by identifying (i) one or more third-party immutable digital testimonies, (ii) one or more cameras, or (iii) a combination of both (i) and (ii), each along a path taken by a user during a user-predetermined time prior to an immutable digital testimony creation; identifying and visually indicating a location of (i) the one or more third-party immutable digital testimonies, (ii) the one or more cameras, or (iii) a combination of both (i) and (ii); sending the map of the immutable digital testimony trail to a device of a creator of immutable digital testimony, the map to be associated with the immutable digital testimony local media stored on the device of the creator; and recording on the immutable digital testimony authentication blockchain, by the immutable digital testimony orchestrator: the immutable digital testimony trail associated with (i) an immutable digital testimony identifier (ID) and (ii) immutable digital testimony geodata, each associated with the creator of the immutable digital testimony local media, and one or more of an immutable digital testimony identifier corresponding to (i) the one or more third-party immutable digital testimonies, (ii) the one or more cameras, or (iii) both (i) and (ii), correspondingly.

[0026] In some embodiments, the method further involves presenting a thumbnail of a specific immutable digital testimony ID on an immutable digital testimony gallery for the creator of the immutable digital testimony; and presenting the immutable digital testimony trail as part of metadata of the immutable digital testimony local media in the immutable digital testimony gallery, wherein any one or a combination of (i) the one or more third-party immutable digital testimonies and (ii) the one or more cameras, each of which allow for the requesting of an immutable digital testimony view are identified with one or more icons, correspondingly; initiating an immutable digital testimony viewing or immutable digital testimony sharing request for a corresponding immutable digital testimony by selecting any of the one or more icons.

[0027] In some embodiments, there is provided a system having one or more processors communicatively coupled to a plurality of sensors, and a non-transitory memory, the non-transitory memory storing instructions that when executed by the one or more processor, control the system to create and view an immutable digital testimony, comprising: a first layer, a second layer and a third layer.

[0028] The first layer operates on a device having an immutable digital testimony device identifier ID, wherein the immutable digital testimony device ID is a unique identifier of the creator device, the device configured to: activate an immutable digital testimony creation operation; authenticate a creator of an immutable digital testimony based on an immutable digital testimony creator ID, wherein the immutable digital testimony creator ID is a unique user identification created upon registering with an immutable digital testimony network; capture the immutable digital testimony device ID of the device used to create the immutable digital testimony; upon authentication of the creator, initiate an immutable digital testimony capture operation to: activate one or more cameras associated with the device and cause the device to generate immutable digital testimony stream media; hash one or more frames of the immutable digital testimony stream media for the one or more cameras by applying a selected immutable digital testimony hashing scheme from a plurality of hashing schemes, thereby creating one or more immutable digital testimony stream hashes; form a live stream hash from the one or more immutable digital testimony stream hashes; capture immutable digital testimony metadata to be recorded on an immutable digital testimony authentication blockchain; transmit the immutable digital testimony stream media and the selected immutable digital testimony hashing scheme to a cloud server, causing the selected immutable digital testimony hashing scheme and the immutable digital testimony stream media to be stored on the cloud server as immutable digital testimony cloud media; retrieve one or more immutable digital testimony addresses, each of the one or more immutable digital testimony addresses being one or more cloud addresses on the cloud server, one or more distributed file system DFS addresses on the DFS, or a combination of i and ii, correspondingly, wherein each of the one or more immutable digital testimony addresses corresponds to a location at which the immutable digital testimony cloud media is saved, the immutable digital testimony cloud media being at least a portion of the immutable digital testimony stream media; upon completion of the transmission: encrypt the immutable digital testimony live stream media thereby creating encrypted immutable digital testimony live stream media, the encrypted immutable digital testimony live stream media being at least a portion of the immutable digital testimony stream media, send to the DFS the encrypted immutable digital testimony live stream media to save the encrypted immutable digital testimony live stream media, and receive from the DFS one or more immutable digital testimony DFS addresses corresponding to the location on the DFS where the immutable digital testimony stream media is saved, wherein each of the one or more immutable digital testimony DFS addresses corresponds to a location at which the encrypted immutable digital testimony live stream media has been saved; and save the immutable digital testimony stream media on the device, to create immutable digital testimony local media.

[0029] In some embodiments, the device is further configured to: select one or more frames of the immutable digital testimony local media corresponding to the same one or more frames of the immutable digital testimony stream media; hash one or more frames of the selected one or more frames of the immutable digital testimony local media using the selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony local hashes; form a local symbiotic hash from the one or more immutable digital testimony local hashes; send to a smart contract, the local symbiotic hash for validating against the live stream hash; and send to the smart contract, the live stream hash, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, and the one or more immutable digital testimony cloud addresses.

[0030] In some embodiments the system further involves a third layer configured to: select one or more frames of the immutable digital testimony cloud media corresponding to the same one or more frames of the immutable digital testimony stream media; hash one or more frames of the selected one or more frames of the immutable digital testimony cloud media using the selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony cloud hashes; form a cloud media hash from the one or more immutable digital testimony cloud hashes; and send, from the cloud server to the smart contract, the cloud media hash for validating against the live stream hash.

[0031] In some embodiments, the third layer is configured to: create, by the smart contract, a creator token by combining the immutable digital testimony creator ID and the immutable digital testimony device ID; create, by the smart contract, an immutable digital testimony identifier ID by hashing the creator token with the live stream hash; and compare, by the smart contract, the live stream hash, the local symbiotic hash, and the cloud media hash, and if a) the live stream hash mirrors the local symbiotic hash and b) the live stream hash mirrors the cloud media hash: create, by the smart contract, an immutable digital testimony hash by hashing the immutable digital testimony ID, the one or more immutable digital testimony cloud addresses, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, and the live stream hash, record, by the smart contract: the immutable digital testimony ID, the one or more immutable digital testimony cloud addresses, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, the immutable digital testimony hash, the live stream hash on the immutable digital testimony authentication blockchain, and the user permissions on an immutable digital testimony authentication blockchain, send the user permissions to an immutable digital testimony orchestrator to create, based on the user permissions, a log entry indicating whether a corresponding immutable digital testimony creator ID has any rights over the immutable digital testimony cloud media that is associated with a specific immutable digital testimony ID, and record, by the smart contract, the immutable digital testimony ID and immutable digital testimony hash on a public blockchain.

[0032] In some embodiments, the third layer is further configured to: if a) the live stream hash and the local symbiotic hash do not match or b) the live stream hash and a cloud media hash do not

match: create an alternate immutable digital testimony by saving the immutable digital testimony local media to the DFS and the cloud server and retrieving the one or more alternate immutable digital testimony DFS addresses and the one or more alternate immutable digital testimony cloud addresses, correspondingly, and provide a notification, by the smart contract, that the immutable digital testimony stream media was corrupted during the transmission or storage process.
[0033] In some embodiments, the device is further configured to: send the immutable digital testimony metadata, the live stream hash, the local symbiotic hash, the one or more alternate immutable digital testimony cloud addresses, and the one or more alternate immutable digital testimony DFS addresses, to the smart contract.
[0034] In some embodiments, the smart contract is further configured to: receive the alternate immutable digital testimony metadata, the live stream hash, the local symbiotic hash, and the one or more alternate immutable digital testimony DFS addresses; create an alternate creator token by combining the immutable digital testimony creator ID and the immutable digital testimony device ID; create an immutable digital testimony alternate identifier (ID) by hashing the alternate creator token with the live stream hash and adding ALT at the end of the immutable digital testimony alternate ID; create the alternate immutable digital testimony hash by hashing the immutable digital testimony alternate ID, the immutable digital testimony metadata, the one or more alternate immutable digital testimony cloud addresses, the one or more alternate immutable digital testimony DFS addresses, the local symbiotic hash, and the live stream hash; record the immutable digital testimony alternate ID, the immutable digital testimony metadata, the one or more alternate immutable digital testimony cloud addresses, the one or more alternate immutable digital testimony DFS addresses, the live stream hash, the local symbiotic hash, and the alternate immutable digital testimony hash to the immutable digital testimony authentication blockchain; record user permissions corresponding to the immutable digital testimony alternate ID on the authentication blockchain and sending the user permissions to an immutable digital testimony orchestrator to update log entries of the immutable digital testimony orchestrator; and record the alternate immutable digital testimony ID and alternate immutable digital testimony hash on the public blockchain.
[0035] In some embodiments, the immutable digital testimony orchestrator is configured to: receive user viewing permissions to an immutable digital testimony corresponding to the immutable digital testimony alternate ID, and create a local ledger of permission details based on the user permissions.
[0036] In some embodiments, the system further comprising: a requestor device configured to: authenticate a third-party immutable digital testimony requestor; initiate from an immutable digital testimony creator, a third-party request, the third-party request being any one or more of: an immutable digital testimony request, requesting to create and share a new immutable digital testimony, an immutable digital testimony view request, requesting permission to view an existing immutable digital testimony of the creator, or an immutable digital testimony share request, requesting permission to share an existing immutable digital testimony of the creator with another immutable digital testimony network user; create the third-party request by combining any one or a combination of: the third-party immutable digital testimony requestor ID, a phone number of an immutable digital creator, and an immutable digital testimony ID, and send the third-party request to an immutable digital testimony orchestrator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0037] The features and advantages of the example embodiments of the invention presented herein will become more apparent from the detailed description set forth below when taken in conjunction

with the following drawings.
[0038] FIG. **1** illustrates an example network for creating and viewing an immutable digital testimony, according to an example embodiment.
[0039] FIG. **2** depicts a device with stored technical processes that collaborate to operate as an immutable digital testimony application executable by the device, according to an example embodiment.
[0040] FIG. **3** illustrates a system-flow diagram of an immutable digital testimony creation process, according to an example embodiment.
[0041] FIG. **4** illustrates a third-party request process for requesting an immutable digital testimony, according to an example embodiment.
[0042] FIG. **5** illustrates an immutable digital viewing process, according to an example embodiment.
[0043] FIG. **6** illustrates a process for creating an immutable digital testimony trail, according to an example embodiment.
[0044] FIG. **7** illustrates an immutable digital testimony authentication process, according to an example embodiment.

DETAILED DESCRIPTION

[0045] The example embodiments of the invention presented herein are directed to methods, systems and computer program products for performing signal processing on various signals to provide immutable digital testimony.
[0046] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art of this disclosure. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well known functions or constructions may not be described in detail for brevity or clarity.
[0047] “Immutable digital testimony”, “IDT”, “immutable digital witness” or “IDW” as used herein generally mean digital evidence, testimony, or information that cannot be changed, altered, or tampered with. It ensures the integrity and authenticity of such evidence, testimony, or information by using the techniques described herein to create a verifiable and unchangeable record.
[0048] FIG. **1** illustrates an example network **100** for creating and viewing an immutable digital testimony, according to an example embodiment. As shown in FIG. **1**, in an example implementation, network **100** is comprised of three technical layers that together enable the creation and viewing of immutable digital testimony: a first layer **1100**, also referred to as a witness layer **1100**, a second layer **1200**, also referred to as an orchestrator layer **1200**, and a third layer **1300**, also referred to as an immutable digital testimony (IDT) layer **1300**. The first layer **1100**, the second layer **1200**, and the third layer **1300** are each a technical layer within a layered model that defines how data is transmitted and processed over the network **100**. The layers of network **100** work together to facilitate the seamless creation, storage, and streaming of immutable digital testimonies (IDTs).
[0049] In some embodiments, first layer **1100** operates on one or more devices **1110** capable of communicating on network **100**, where a device **1110** has an installed on non-transitory medium an application that, when executed by the device **1110**, causes the device **1110** to perform one or more of the processes described herein. In an example implementation, a device **1110** is a mobile device having one or more processors and a non-transitory memory for storing the application. In some implementations, a device **1110** is a camera enabled to communicate on network **100**. In some implementations, the camera includes one or more processors and a non-transitory memory for storing the application. In some implementations, the camera is communicatively coupled to

another device (e.g., a computer, an edge device, and the like) having one or more processors and a non-transitory memory for storing the application. In some embodiments, the one or more devices **1110** include a network of cameras enabled to communicate on network **100**.

[0050] In some implementations, one or more devices **1110** capture media including audio content and/or video content and communicate the media on network **100**.

[0051] Second layer **1200** includes an immutable digital testimony (IDT) orchestrator **1210**. Generally, IDT orchestrator **1210** manages user permissions, user-to-user messaging, payment processing, and/or system administration. In some embodiments, the second layer **1200** runs on a cloud which can deliver various services over the internet, such as storage, processing power, databases, networking, software, and analytics. In some embodiments, the second layer **1200** runs on a blockchain infrastructure (referred to simply as a "blockchain"). Advantageously, the processes, data, or functionalities associated with embodiments described herein utilize the blockchain, which offer benefits like enhanced security, transparency, and immutability.

[0052] The third layer **1300** generally operates to secure authentication and validation keys, execute smart contracts, manage user permissions, host media content, and authorize streaming of the immutable digital testimony. A "smart contract" as used herein generally means a self-executing contract with the terms of an agreement directly written into code. In some embodiments these contracts automatically enforce and execute the terms and conditions when predefined conditions are met, without the need for intermediaries. The smart contracts described herein run on a blockchain, ensuring transparency, security, and immutability. This enables automated, reliable, and transparent transactions and agreements in the applications described herein. In some embodiments, the smart contract is in the IDT layer **1300**. An "immutable digital testimony (IDT) authentication blockchain" as used herein generally refers to a blockchain that contains immutable digital testimony related information that is used to validate and authenticate an immutable digital testimony and provide an immutable log and an audit trail related to the creation, viewing, and sharing of immutable digital testimonies. Referring to FIG. **1**, in an example implementation, the smart contract resides on IDT authentication blockchain **1330** and operates to validate the integrity of immutable digital testimony (IDT) stream media and record the immutable digital testimony and related information to applicable blockchains (e.g., the IDT authentication blockchain **1330** and a public blockchain **1340**).

[0053] "Immutable digital testimony (IDT) stream media" as used herein generally refers to media that is transmitted (e.g., streamed, downloaded, or uploaded, or otherwise communicate, as applicable) by a device **1110** (e.g., a mobile device or camera) on network **100**. In an example implementation, the IDT stream media is transmitted to IDT cloud **1310**.

[0054] In some embodiments, immutable digital testimony (IDT) cloud **1310** is a cloud-based media storage system where the IDT stream media and related information are received and stored. IDT stream media and related information received and stored in IDT cloud **1310** becomes immutable digital testimony (IDT) cloud media. In other words, IDT cloud media is media that is stored on the IDT cloud. It should be understood that the IDT cloud and any blockchain described herein is non-transitory media.

[0055] A "distributed file system (DFS)" as used herein generally means a decentralized file system, including but not limited to the Interplanetary File System, where the immutable digital testimony related data might be stored for validation or media recovery. A DFS, in some embodiments, is a file system that enables a client device (referred to sometimes simply as a "client") to access file storage from multiple hosts through a computer network as if the client device was accessing local storage. Files are spread across multiple storage servers and in multiple locations, enabling client devices to share data and storage resources.

[0056] "Immutable digital testimony (IDT) authentication" as used herein generally means a request by a creator or user of immutable digital testimony (referred to as "an immutable digital testimony (IDT) creator" or simply "creator") to send an authentication of one of its immutable digital testimonies to a specific user (e.g., to a device of the specific user).

[0057] “Third-party immutable digital testimony (IDT)” as used herein generally means immutable digital testimony created by an entity other than the immediate user or creator, including another user of, or other device on, the IDT network **100**, such as immutable digital testimony cameras, along an immutable digital testimony trail (described below). Such third-party immutable digital testimony is thus created by entities outside of an immediate user or creator.

[0058] Referring to FIG. **1**, in some embodiments, the third layer **1300** is deployed on a combination of IDT cloud **1310**, DFS **1320**, IDT authentication blockchain **1330** and a public blockchain **1340**.

[0059] In an example implementation, a device **1110** includes a plurality of cameras, such as a front camera, a back camera, or any other camera interconnected to the device **1110**, configured to capture audio and/or visual content to be used as IDT stream media. The IDT stream media corresponds to the live media while it is in transit to the IDT cloud **1310**, where it will be stored and become IDT cloud media.

[0060] FIG. **2** depicts a device **1110** with stored technical processes **200** that collaborate to operate as an immutable digital testimony application **120** executable by the device, according to an example embodiment. In some embodiments, device **1110** includes a user input device **106**, a display device **108**, a data communication device **110**, a processing device **112**, one or more media capture devices **116**, and a memory device **118**. Aspects of the technical processes **200** are performed locally at the device **1110** via the IDT application **120**. Referring also to FIG. **1**, certain related processes are performed on the second layer **1200** or the third layer **1300** of network **100**. In some embodiments, the IDT application **120** executed on the device **1110** in communication with the IDT cloud **1310**, DFS **1320**, IDT authentication blockchain **1330** and/or public blockchain **1340** over network **100**.

[0061] A user input device **106** of device **1110** operates to receive a user input from a user for controlling the device **1110**. The user input can include a manual input and/or a voice input, among others. In some embodiments, the user input device **106** includes one or more buttons, keys, touch levers, switches, and/or other mechanical input devices for receiving the input. The user input device **106** can include a touch screen or a gesture input device. In some embodiments, the user input device **106** can detect sounds including the voice input such as a voice of a user (e.g., an utterance) for controlling aspects of a technical processes via the device **1110**.

[0062] In some embodiments, a display device **108** is provided that operates to display a graphical user interface that displays information for interacting with the device **1110**. Examples of such information include an immutable digital testimony trail or immutable digital testimony gallery, both described in more detail below. In some embodiments, the display device **108** is configured as a touch sensitive display and includes the user input device **106** for receiving input from a selector (e.g., a finger, stylus etc.) controlled by a user. In some embodiments, therefore, the display device **108** operates as both a display device and a user input device.

[0063] The data communication device **110** operates to enable the device **1110** to communicate with one or more computing devices over one or more networks, such as network **100**. For example, the data communication device **110** is configured to communicate with the IDT cloud **1310**, DFS **1320**, IDT authentication blockchain **1330** and/or public blockchain **1340** over network **100** and receive notifications from the IDT cloud **1310**, DFS **1320**, IDT authentication blockchain **1330** and/or public blockchain **1340**, among other components, via network **100**. The data communication device **110** can be a network interface of various types which connects the device **1110** to the network **100**.

[0064] The processing device **112**, in some embodiments, comprises one or more central processing units (CPU). In other embodiments, the processing device **112** additionally or alternatively includes one or more digital signal processors, graphical processing units (GPUs), field-programmable gate arrays, or other electronic circuits.

[0065] The media capture device **116**, in some embodiments, is one or more cameras integrated with the device **1110**. The media capture device **116** is configured to capture media such as audio and/or video. It should be understood that video can include one or more still images as part of its content.
[0066] The memory device **118** includes at least some form of non-transitory computer-readable media. Non-transitory computer-readable media includes any available media that can be accessed by the device **1110**, such as volatile and nonvolatile, removable and non-removable media implemented in any device configured to store information such as computer readable instructions, data structures, program modules, or other data. Memory device **118** can also include, but is not limited to, random access memory, read only memory, electrically erasable programmable read only memory, flash memory and other memory technology, compact disc read only memory, blue ray discs, digital versatile discs or other optical storage, magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed by the device **1110** in a non-transitory manner.
[0067] The memory device **118** operates to store data and instructions. In some embodiments, the memory device **118** stores instructions for an IDT application **120**. In some examples, one or more methods described herein can be implemented as instructions stored in one or more memory devices on one or more computers that, when executed by one or more processors, cause the processors to perform one or more operations described herein.
[0068] In some examples, network **100** includes a computer network, an enterprise intranet, the Internet, a LAN, a Wide Area Network (WAN), wireless transmission mediums, wired transmission mediums, satellite transmission mediums, other networks, and combinations thereof. Although network **100** is shown as a single network in FIG. **1**, this is shown as an example and the various communications described herein can occur over the same network or a number of different networks.
[0069] The technical processes **200** that collaborate to operate as an immutable digital testimony (IDT) application **120** include an immutable digital testimony (IDT) creation process **2100**, an immutable digital testimony (IDT) request process **2200**, an immutable digital testimony (IDT) viewer process **2300**, an immutable digital testimony (IDT) trail process **2400**, an immutable digital testimony (IDT) authentication process **2500**, an immutable digital testimony (IDT) gallery process **2600**, and immutable digital testimony (IDT) administrative functions **2700**.
[0070] These technical processes **200**, in some embodiments, are in the form of instructions that, when executed by one or more processors of a device **1110**, cause the device **1110** to perform the applicable operations which are described in more detail herein.
[0071] The IDT creation process **2100** generally is a process that creates an immutable digital testimony.
[0072] The IDT request process **2200** generally is a process for handling requests for an immutable digital testimony. The IDT request process **2200** can be performed according to different sub-processes, each for a different purpose: a third-party immutable digital testimony (IDT) request process **2210**, an immutable digital testimony (IDT) viewing process **2220**, and an immutable digital testimony (IDT) sharing request process **2230**.
[0073] The third-party IDT request process **2210** is a request process initiated by a first client (referred to as an immutable digital testimony (IDT) requestor or simply a requestor) to a second client (referred to as an immutable digital testimony (IDT) creator or simply creator) for initiating a new immutable digital testimony.
[0074] The IDT viewing process **2220** generally is a process that receives a request from a requestor initiating a request for an immutable digital testimony (also referred to as an immutable digital testimony (IDT) requestor or simply requestor) to the person who has created one or more immutable digital testimonies (also referred to as an immutable digital testimony (IDT) creator or simply creator) to grant permission to view at least one of the immutable digital testimonies created

by the person who created the immutable digital testimony.

[0075] The IDT share process **2230** generally is a request via a device operated by a person making the request (also referred to as an immutable digital testimony (IDT) requestor or simply requestor) for a creator who has created one or more immutable digital testimonies (also referred to as an immutable digital testimony (IDT) creator or simply creator) to grant permission, via a device, to share with another user of the network **100** at least one of the immutable digital testimonies created by the creator who created the immutable digital testimony(ies).

[0076] The IDT viewer process **2300** generally is a process that performs operations for transmitting (e.g., streaming, downloading, uploading, or otherwise communicating, as applicable) and viewing of immutable digital testimonies.

[0077] The IDT trail process **2400** generally is a process that provides (e.g., by presenting via a display of a device) a map identifying a trail along the path a person followed while creating an immutable digital testimony via a device, within a user-determined timeframe. In some embodiments, all third-party immutable digital testimonies that are captured by other persons who created immutable digital testimonies using, e.g., mobile devices, or by cameras enabled to constantly stream to the network **100**, are also provided on the map. In an example implementation, each immutable digital testimony has a unique corresponding immutable digital testimony trail which is constructed using immutable digital testimony geodata. "Immutable digital testimony (IDT) geodata" generally means a geolocation audit-trail that identifies the whereabouts of a person who created the immutable digital testimony through one or more devices during a user defined timeframe (e.g., a 60-minutes period). In an example implementation, this can include immutable digital testimony created through one or more devices during a user defined timeframe prior to initiating an immutable digital testimony capture.

[0078] The IDT authentication process **2500** generally is a process for authenticating an immutable digital testimony.

[0079] The IDT gallery process **2600** generally is a process that displays one or more immutable digital testimonies including, for example, any immutable digital testimony created through a self-initiated immutable digital testimony process or an IDT request process **2200** (e.g., a via third-party IDT request process **2210**, IDT viewing process **2220**, or IDT sharing request process **2230**).

[0080] The IDT administrative functions **2700** generally perform various operations related to the administration of immutable digital testimonies.

[0081] FIG. **3** illustrates a system-flow diagram of an IDT creation process **2100**, according to an example embodiment. In some embodiments, the IDT creation process **2100** is performed by the witness layer **1100** (also referred to as the first layer **1100**).

[0082] In an example implementation, the IDT creation process **2100** begins when an immutable digital testimony creation feature is activated on a device, such as device **1110**, as illustrated by activate IDT creation operation **2102**. In some embodiments, the IDT creation process **2100** is activated on the device by a user associated with an IDT creator identifier (ID). In some embodiments, the IDT creator ID is a unique user identification that is created upon the IDT creator registering with the network **100**.

[0083] The device performs a user authentication operation **3100** to authenticate the user. In an example implementation, user authentication operation **3100** performs capturing a device ID and authenticating a user based on the device ID and an IDT creator ID associated with the user. The device ID, in some embodiments, is a unique device identification number, such as the International Mobile Equipment Identity ("IMEI") or similar unique identifier, for the device used to capture the immutable digital testimony.

[0084] Upon successful authentication, the device initiates an immutable digital testimony (IDT) capture process **3110**. IDT capture process **3110** involves activating available cameras (e.g., one or more cameras) to create IDT stream media, as illustrated by activate cameras operation **3120**. The device, in turn, performs a form live stream hash operation **3130**. In an example implementation, form live stream hash operation **3130** generates a live stream hash by, for each available camera, hashing each frame of a related IDT stream media **3131** using a selected immutable digital testimony (IDT) frame hashing scheme **3132**.

[0085] An IDT hashing scheme generally means any one of a variety of schemes for how to hash frames, including among others, for example, dividing individual frames of video into multiple pieces, such as 1 equal part, 2 equal parts, 4 equal parts, 3 unequal parts, into 4 triangles, hashing alternating frames rather than each frame, hashing partial frames rather than complete frames, etc. By assigning an IDT hashing scheme to be used for hashing an IDT stream media, it becomes more difficult for hackers to reverse-engineer the immutable digital testimony validation process and manipulate the media that is stored in the IDT cloud **1310** or DFS **1320**. A selected IDT hashing scheme becomes the IDT frame hashing scheme for that immutable digital testimony.

[0086] Once the live stream is complete, the IDT stream hashes from each camera are combined or hashed to form the live stream hash **3133**.

[0087] The device also performs an immutable digital testimony (IDT) metadata capture operation **3140** to capture immutable digital testimony (IDT) metadata. In some embodiments, the IDT metadata includes various truth attestation factors. In an example implementation, truth attestation factors include one or more immutable digital testimony (IDT) creator identifiers (IDs) **3141**, one or more device identifiers (IDs) **3142**, media metadata **3143**, location metadata (**3144**), an immutable digital testimony (IDT) frame hashing scheme **3145**, and an immutable digital testimony (IDT) source code identifier (ID) **3146**.

[0088] In an example implementation, immutable digital testimony creator identifiers **3141** include: an immutable digital testimony creator ID of a creator, an immutable digital testimony ID method, a name of the user, an e-mail address of the user and one or more validated government identifiers (IDs) of the user. An IDT ID method as used herein generally means the method by which the IDT creator ID was captured. In an example implementation, IDT ID method obtains the IDT creator IDT by using biometric information. In another example implementation, IDT ID method obtains the IDT creator IDT by using manual entry through a device (e.g., user input device **106** of device **1110**).

[0089] In an example implementation, device identifiers **3142** include: an immutable digital testimony (IDT) device identifier (ID), a phone number (if applicable), a subscriber identity module (SIM) card information (if applicable), an internet protocol (IP) address of the device, a date and time as reported by the device, and a battery level as reported by the device.

[0090] In an example implementation, media metadata **3143** includes: Light Detection and Ranging (LiDAR) data for each available camera of a device.

[0091] In an example implementation, location metadata **3144** includes latitude, longitude, elevation, and other location identifiers of the device (with the specific IDT device ID that was used when the IDT capture process **3110** was initiated, IDT geodata, and compass and heading of the device associated with the IDT device ID.

[0092] It should be understood that in alternative implementations other combinations of metadata can be within the scope of the embodiments herein.

[0093] In an example implementation, an immutable digital testimony (IDT) frame hashing scheme **3145** is one or more corresponding immutable digital testimony (IDT) frame hashing schemes associated with the IDT stream media.

[0094] In an example implementation, immutable digital testimony (IDT) source code identifier **3146** is an identifier of the code used to create an immutable digital testimony with a specific immutable digital testimony identifier (IDT ID). In some embodiments, IDT source code that was used to create a specific immutable digital testimony is immutable. Being immutable preserves a record of the source code which can be reviewed in the future to ensure there was no malware or manipulation at the moment of the immutable digital testimony creation.

[0095] An IDT ID, in some embodiments, is a unique identifier assigned to every immutable digital

testimony which is created by hashing a creator token with a live stream hash. In an example implementation, a live stream hash is a hash or combination of all the IDT stream hashes that are used to ensure that corresponding live streamed media was not altered or swapped during a transmission.
[0096] Simultaneously with the formation of the live stream hash and the capturing of the immutable digital testimony metadata, the device operates to perform transmitting and saving operation **3147** to transmit and save the IDT stream media and a selected IDT frame hashing scheme to the immutable digital testimony cloud. The selected IDT frame hashing scheme, in some embodiments, is a selected IDT hashing scheme used to hash the IDT stream media prior to it being transmitted to the IDT cloud **1310**.
[0097] It should be understood that "simultaneously" can mean substantially simultaneously, and either the formation of the live stream hash or the capturing of the immutable digital testimony metadata can occur in different orders, and still be within the same scope of the embodiments herein.
[0098] A live stream is transmitted (e.g., streamed, uploaded, downloaded or otherwise communicated by or from a device, such as device **1110**, as applicable) to the IDT cloud **1310** and as illustrated by communication operation **39100**. In response, IDT cloud **1310** responds to the device causing the device to retrieve a corresponding IDT cloud address, as illustrated by communication operation **39110**. An "immutable digital testimony (IDT) cloud address" as used herein generally means the address on the IDT cloud **1310** where IDT stream media and related information are saved, thus becoming the immutable digital testimony cloud media.
[0099] Live stream media generally refers to multimedia content that is broadcast in real-time over the internet or a network. This includes live media feeds (including live video and audio feeds) that are transmitted as they are being recorded. Upon completion of the immutable digital testimony transmission, the live stream media which has been cached on the device is encrypted by an encryption operation **3148** and sent to the DFS **1320** as illustrated by communication operation **39105**. In response, DFS **1320** returns to the device a corresponding immutable digital testimony (IDT) DFS address as illustrated by communication operation **39115**. An "IDT DFS address" as used herein generally means the DFS address of where a live stream media is stored as described herein.
[0100] "Immutable digital testimony (IDT) local media" as used herein generally means the IDT stream media that is cached and stored locally on a device, such as device **1110**. In some embodiments, for each camera, the IDT stream media on the device is saved to create IDT local media, as illustrated by local media stored operation **3149**.
[0101] An "immutable digital testimony (IDT) local hash" as used herein generally means a hash that is created for each frame of an IDT local media using a corresponding IDT frame hashing scheme.
[0102] A "local symbiotic hash" as used herein generally means a hash or combination of the IDT local hashes that is/are used to compare the IDT stream media and corresponding IDT local media. In some embodiments, the device (e.g., device **1110**) performs a local symbiotic hash formation operation **3150** to create a local symbiotic hash **3153** by selecting frames of the IDT local media **3151** and hashing each frame of the IDT local media **3151** to create a so-called "immutable digital testimony (IDT) local hash", using the IDT frame hashing scheme **3152** that was used during the creation of the IDT stream media and to form the IDT media hash. In turn, the device performs a combining or hashing operation to combine or hash the IDT local hashes to form a local symbiotic hash **3153**.
[0103] The local symbiotic hash is then sent to a smart contract as illustrated by communication operation **39020**.
[0104] In turn, device **1110** performs a send operation **3160** in conjunction with communication operation **39030** to send the IDT metadata, the live stream hash, the IDT DFS address, and the IDT

cloud address to the smart contract.

[0105] In some embodiments, a method for creating an immutable digital testimony involves an activate IDT creation operation **2102** to perform activating an immutable digital testimony creation operation on a device **1110** having an immutable digital testimony device identifier (ID), where the immutable digital testimony device ID is a unique identifier of the device **1110**. The method further involves a user authentication operation **3100** to perform authenticating a creator of an immutable digital testimony based on an immutable digital testimony creator ID, where the immutable digital testimony creator ID is a unique user identification created upon registering with an immutable digital testimony network. In turn, the method performs an immutable digital testimony (IDT) capture process **3110** to perform capturing the immutable digital testimony device ID of the device used to create the immutable digital testimony, and upon authentication of the creator, initiating by the device, an immutable digital testimony capture operation.

[0106] In some embodiments, the IDT capture operation **3110** includes: an activate cameras operation **3120** that performs activating one or more cameras associated with the device and causing the device to generate immutable digital testimony stream media, a form live stream hash operation **3130** to perform hashing one or more frames of the immutable digital testimony stream media for the one or more cameras by applying a selected immutable digital testimony hashing scheme from a plurality of hashing schemes, thereby forming one or more immutable digital testimony stream hashes and forming a live stream hash from the one or more immutable digital testimony stream hashes.

[0107] The method further involves an immutable digital testimony (IDT) metadata capture operation **3140** to perform capturing immutable digital testimony metadata to be recorded on an immutable digital testimony authentication blockchain, transmit and communication operations **3147**, **39100** to perform transmitting the immutable digital testimony stream media and the selected immutable digital testimony hashing scheme by the device to a cloud server, causing the selected immutable digital testimony hashing scheme and the immutable digital testimony stream media to be stored on the cloud server as immutable digital testimony cloud media.

[0108] In turn, the method involves a communication operation **39110** to perform retrieving one or more immutable digital testimony addresses, each of the one or more immutable digital testimony addresses being (i) one or more cloud addresses on the cloud server, (ii) one or more distributed file system (DFS) addresses on the DFS, or (iii) a combination of (i) and (ii), correspondingly, wherein each of the one or more immutable digital testimony addresses corresponds to (i) a location at which the immutable digital testimony cloud media is saved, the immutable digital testimony cloud media being at least a portion of the immutable digital testimony stream media, or (ii) a location allocated for saving the immutable digital testimony cloud media which is at least a portion of the immutable digital testimony stream media. Upon completion of the transmitting the method involves: an encryption operation **3148** to perform encrypting the immutable digital testimony live stream media, thereby creating encrypted immutable digital testimony live stream media, the encrypted immutable digital testimony live stream media being at least a portion of the immutable digital testimony stream media, sending to the DFS the encrypted immutable digital testimony live stream media by a communication operation **39105** to save the encrypted immutable digital testimony live stream media. In turn, the method involves a receiving operation **39015** that causes receiving one or more immutable digital testimony DFS addresses, where each of the one or more immutable digital testimony DFS addresses corresponds to a location at which the encrypted immutable digital testimony live stream media has been saved. A local media stored operation **3149**, in turn, performs saving the immutable digital testimony stream media on the device to create immutable digital testimony local media.

[0109] The immutable digital testimony metadata, in some embodiments, further comprises any one or a combination of: biometric data of an immutable digital testimony creator, manual entry data of an immutable digital testimony creator, a name of an immutable digital testimony creator, an e-mail address of an immutable digital testimony creator, a validated government identifier of the immutable digital testimony creator, a phone number of the device, SIM card information of the device, a date and time as reported by the device, a battery level as reported by the device, LiDAR data for each of the one or more cameras associated with the device, a location identifier of the device when a camera capture operation was initiated, and other data related to the user, device, location, and media used to create the immutable digital testimony.

[0110] In some embodiments, forming the live stream hash is performed by combining or hashing the one or more immutable digital testimony stream hashes. In an example implementation, the combining is performed using a hashing scheme (e.g., Merkle Root).

[0111] In some embodiments, the method further involves a local symbiotic hash formation operation **3150** to perform selecting one or more frames of the immutable digital testimony local media corresponding to the same one or more frames of the immutable digital testimony stream media, hashing one or more frames of the selected one or more frames of the immutable digital testimony local media using the selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony local hashes. In turn, the local symbiotic hash creation operation **3150** performs forming a local symbiotic hash from the one or more immutable digital testimony local hashes. A communication operation **39020** performs sending, by the device to a smart contract, the local symbiotic hash for validating against the live stream hash. A send operation **3160** and communication operation **39030** perform sending, by the device to the smart contract, the live stream hash, the immutable digital testimony metadata, the one or more immutable digital testimony DFS addresses, and the one or more immutable digital testimony cloud addresses.

[0112] In some embodiments, forming the local symbiotic hash is performed by combining or hashing the one or more immutable digital testimony local hashes. In an example implementation, the combining is performed using a hashing scheme (e.g., Merkle Root).

[0113] Referring still to FIG. **3**, in some embodiments, IDT cloud **1310** is used to compare the IDT stream media received and IDT cloud media stored on the IDT cloud **1310** to ensure that the media that was streamed from the device was not altered or manipulated by performing a cloud media hash formation operation **3310** to create a cloud media hash. As explained above, in some embodiments, IDT cloud **1310** is on IDT layer **1300**.

[0114] A “cloud media hash” as used herein generally means a hash or combination of all immutable digital testimony cloud hashes that are used to compare the IDT stream media and the immutable digital testimony cloud media.

[0115] An “immutable digital testimony (IDT) cloud hash” generally refers to hashes that are formed for each frame of the IDT cloud media, using the corresponding IDT frame hashing scheme.

[0116] Cloud media hash formation operation **3310** performs selecting one or more frames of the IDT cloud media corresponding to the same one or more frames of the IDT stream media. In turn, a hashing operation performs hashing one or more frames of the selected one or more frames of the IDT cloud media using the selected IDT frame hashing scheme to create one or more IDT cloud hashes **3312**. In turn, cloud media hash formation operation **3310** performs forming a cloud media hash **3313** from the one or more IDT cloud hashes **3312**.

[0117] In some embodiments, forming the cloud media hash is performed by combining or hashing the one or more immutable digital testimony cloud hashes. In an example implementation, the combining is performed using a hashing scheme (e.g., Merkle Root).

[0118] In turn, a communication operation **39040** performs sending, by the cloud server to the smart contract, the cloud media hash for validating against the live stream hash.

[0119] A creator token, generally, is the combination of an IDT user ID and an IDT device ID. Particularly, a smart contract on the IDT authentication blockchain **1330** performs a create token operation **3331** to create the creator token by combining the IDT creator ID and the IDT device ID.

[0120] The smart contract, in turn, performs a create immutable digital testimony (IDT) identifier (ID) operation **3332** to create an immutable digital testimony (IDT) identifier (ID) by hashing the creator token with the live stream hash. The smart contract, in some embodiments, performs a validation operation **3333** to validate the IDT cloud media to ensure the same was not corrupted during transmission or saving on the IDT cloud **1310** and/or the DFS **1320**, by verifying that there is no difference between the IDT stream media, the IDT local media, and the IDT cloud media. In an example implementation, this is performed by comparing the live stream hash (corresponding to the IDT stream media), the local symbiotic hash (corresponding to the IDT local media), and the cloud media hash (corresponding to the IDT cloud media).
[0121] In some embodiments, if a) the live stream hash mirrors the local symbiotic hash and b) the live stream hash mirrors the cloud media hash as shown by validation operation **3333**, then the smart contract forms the IDT hash by hashing the IDT ID, IDT metadata, IDT cloud address, IDT DFS address, and the live stream hash as shown by form IDT hash operation **3380**.
[0122] The smart contract, in turn, performs a record operation **3334** to record the IDT ID, IDT metadata, IDT cloud address, IDT DFS address, live stream hash, and the IDT hash on the IDT authentication blockchain **1330**.
[0123] In addition, smart contract performs a blockchain recording operation **3335** that performs recording the user permissions on the IDT authentication blockchain **1330** and sending the user permissions to the IDT orchestrator **1210**, as shown by communication operation **39300**. In some embodiments, this causes the IDT orchestrator **1210** to update a permission management system **3210** as illustrated by update operation **3220**. Permission management system **3210**, in turn, creates a log entry indicating the specific IDT creator ID that has full rights (viewing and sharing) over the immutable digital testimony associated with a specific IDT ID.
[0124] If applicable, blockchain recording operation **3335** also causes the smart contract to grant a third-party IDT requestor (e.g., upon initiating the third-party IDT request process **2210**) viewing permissions associated with the specific IDT ID. Blockchain recording operation **3335**, in some embodiments, also performs registering the viewing permission on the IDT authentication blockchain **1330** and the permission management system **3210**. It should be understood that in some implementations, permission management system **3210** is incorporated in IDT orchestrator **1210**.
[0125] In some embodiments, the smart contract performs a recording operation **3336** in conjunction with a communication operation **39400** to record the IDT ID and IDT hash on the public blockchain **1340**.
[0126] The IDT hash that is recorded on the public blockchain **1340** can be used for periodic validation and authentication of IDT authentication blockchain **1330**, e.g., the integrity of private blockchain record is validated using the IDT hash stored as a record on the public blockchain **1340**.
[0127] The IDT DFS address and the corresponding media at that location can, in some embodiments, be used to recreate or reconstruct an immutable digital testimony in case the IDT cloud media is corrupted or altered.
[0128] If a determination is made at validation operation **3333** that there is no perfect match between a) the live stream hash and the local symbiotic hash or b) the live stream hash and the cloud media hash, then the smart contract performs a notification operation to notify the immutable digital testimony application **120** that the immutable digital stream media was corrupted during the upload or storage process, as illustrated by communication operation **39050**.
[0129] Another aspect involves a method for creating a cloud media hash for validating against the live stream hash. The method involves a cloud media hash formation operation **3310** for selecting one or more frames of the IDT cloud media corresponding to the same one or more frames of the IDT stream media. The method further involves hashing one or more frames of the selected one or more frames of the immutable digital testimony cloud media using the selected immutable digital testimony frame hashing scheme, thereby creating one or more IDT cloud hashes **3312**. In turn, the

method performs forming a cloud media hash **3313** from the one or more immutable digital testimony cloud hashes. A communication operation **39040** then performs sending, by the cloud server to the smart contract, the cloud media hash for validating against the live stream hash.

[0130] In some embodiments, forming the cloud media hash **3313** is performed by combining or hashing the one or more immutable digital testimony cloud hashes.

[0131] In some embodiments, the method involves a create token operation **3331** for creating, by the smart contract, a creator token by combining the immutable digital testimony creator ID and the IDT device ID.

[0132] A create immutable digital testimony identifier operation **3332** performs creating, by the smart contract, an immutable digital testimony identifier (ID) by hashing the creator token with the live stream hash. A validation operation **3333** performs comparing, by the smart contract, the live stream hash, the local symbiotic hash, and the cloud media hash. If validation operation **3333** determines a) the live stream hash mirrors the local symbiotic hash and b) the live stream hash mirrors the cloud media hash, then the method further involves a form IDT hash operation **3380** that performs forming, by the smart contract, an IDT hash by hashing the IDT ID, the one or more IDT cloud addresses, the IDT metadata, the one or more IDT DFS addresses, and the live stream hash.

[0133] Blockchain recording operations **3334**, **3335** perform, recording, by the smart contract: the IDT ID, the one or more IDT cloud addresses, the IDT metadata, the one or more IDT DFS addresses, the IDT hash, the live stream hash on the IDT authentication blockchain **1330**, and the user permissions on an IDT authentication blockchain **1330**.

[0134] In some embodiments, a communication operation **39300** performs sending the user permissions to an IDT orchestrator **1210** to create, based on the user permissions, a log entry indicating whether a corresponding IDT creator ID has any rights over the IDT cloud media that is associated with a specific IDT ID, and a recording operation **3336** in conjunction with a communication operation **39400** perform recording, by the smart contract, the immutable digital testimony ID and IDT hash on a public blockchain **1340**.

[0135] In some embodiments, the method further performs granting, by the smart contract to a requestor, viewing permission to an immutable digital testimony corresponding to the IDT ID and registering the viewing permission on an IDT authentication blockchain **1330** and the IDT orchestrator **1210**.

[0136] The method, in some embodiments, performs periodically validating and authenticating the immutable digital testimony authentication blockchain by using the immutable digital testimony hash recorded on the public blockchain.

[0137] “Alternate immutable digital testimony” as used herein generally means the alternate immutable digital testimony that was created using the immutable digital testimony local media to preserve the locally saved version when the smart contract determines that an immutable digital testimony was altered or modified.

[0138] “Alternate immutable digital testimony cloud address” as used herein generally means the immutable digital testimony cloud address of where the alternate immutable digital testimony was saved.

[0139] “Alternate immutable digital testimony DFS address” as used herein generally means the DFS address of where the alternate immutable digital testimony was saved.

[0140] In some embodiments, after being alerted by the smart contract of a potential alteration of an immutable digital testimony media, as shown by validation determining operation **3170**, the device **1110** (i.e., on the witness layer **1100**) executes a create alternate immutable digital testimony (IDT) operation **3180** to create an alternate immutable digital testimony (IDT) by performing a saving operation **3190** that saves the IDT local media to the DFS **1320**, as shown by communication operation **39500**, retrieving the corresponding alternate IDT DFS address, as shown by communication operation **39510**. In addition, a saving operation performs saving the IDT local media to IDT cloud **1310**, as shown by communication operation **39511** and retrieving operation perform retrieving the corresponding alternate IDT cloud address, as shown by communication operation **39512**. In some embodiments, a notification operation **39050** performs providing a notification, by the smart contract, that the IDT stream media was corrupted during the transmission or storage process.

[0141] The alternate IDT is recorded on the DFS **1320** to ensure immutability. The alternate immutable digital testimony also is recorded on the IDT cloud **1310** for future playback.

[0142] In some embodiments, device **1110** performs an alternate immutable digital testimony sending operation **3191** that sends the immutable digital testimony metadata, live stream hash, local symbiotic hash, alternate immutable digital testimony cloud address, and alternate immutable digital testimony DFS address to the smart contract via communication operation **39060**.

[0143] In some embodiments, a method performs a validation determining operation **3170** to determine if a) the live stream hash and the local symbiotic hash do not match or b) the live stream hash and a cloud media hash do not match. If so, a create alternate immutable digital testimony operation **3180** performs creating an alternate immutable digital testimony and a saving operation **3190** saves the IDT local media to the DFS via a communication operation **39500** and to the cloud server via a communication operation **39511** and retrieving the one or more alternate immutable digital testimony DFS addresses via communication operation **39510** and the one or more alternate immutable digital testimony cloud addresses via communication operation **39512**, correspondingly.

[0144] In some embodiments, the method further involves an alternate immutable digital testimony sending operation **3191** to perform sending, by the device, the IDT metadata, the live stream hash, the local symbiotic hash, the one or more alternate IDT cloud addresses, and the one or more alternate IDT DFS addresses, to the smart contract via communication operation **39060**.

[0145] In another embodiment a method involves several steps executed by a smart contract (e.g., on the IDT layer **1300**). A communication operation **39060** performs receiving, by the smart contract, the alternate IDT metadata, the live stream hash, the local symbiotic hash, and the one or more IDT testimony DFS addresses.

[0146] A create creator token operation **3341** performs creating, by the smart contract, an alternate creator token by combining the IDT creator ID and the IDT device ID. In turn, a create immutable digital testimony alternate identifier operation **3342** performs creating, by the smart contract, an immutable digital testimony (IDT) alternate identifier (ID) by hashing the alternate creator token with the live stream hash and adding, for example, “ALT” at the end of the IDT alternate ID. An “immutable digital testimony (IDT) altered identifier (ID)” as used herein generally means a unique identifier assigned to an immutable digital testimony that was altered.

[0147] A form alternate immutable digital testimony (IDT) hash operation **3349** performs creating, by the smart contract, the alternate IDT hash by hashing the IDT alternate ID, the IDT metadata, the one or more alternate IDT cloud addresses, the one or more alternate IDT DFS addresses, the local symbiotic hash, and the live stream hash.

[0148] In turn, a record operation **3343** (also referred to as “an IDT authentication blockchain record operation”) performs recording, by the smart contract, the IDT alternate ID, the IDT metadata, the one or more alternate IDT cloud addresses, the one or more alternate IDT DFS addresses, the live stream hash, the local symbiotic hash, and the IDT testimony hash to the IDT authentication blockchain **1330**.

[0149] Another record operation **3344** performs recording, by the smart contract, user permissions corresponding to the IDT alternate ID on the IDT authentication blockchain **1330** and a communication operation **39600** performs sending the user permissions to an IDT orchestrator **1210** to update log entries of the IDT orchestrator **1210** (also referred to as “an IDT orchestrator record operation”). Another record operation **3345** performs recording, by the smart contract, the alternate immutable digital testimony ID and alternate IDT hash on the public blockchain **1340** in conjunction with communication operation **39700** (also referred to as “a public blockchain record operation").

[0150] In some embodiments, the IDT orchestrator **1210** (e.g., in orchestrator layer **1200**) receives the user permissions via communication operation **39600** and, in turn, performs an update operation **3220** to update the IDT orchestrator's permission management system **3210** on the orchestrator layer **1200**, creating a local ledger of permission details. IDT orchestrator **1210**, in some embodiments, also performs a create operation **3230** to create an immutable digital testimony trail, described below in more detail.

[0151] FIG. **6** illustrates a process **600** for creating an immutable digital testimony (IDT) trail **6200**, according to an example embodiment. Generally, each immutable digital testimony can have an IDT trail associated with it. To the extent applicable, each immutable digital testimony can also have associated with it related third-party immutable digital testimonies. The IDT orchestrator **1210** operates to identify all third-party immutable digital testimonies (IDTs) and devices (e.g., mobile devices, cameras, etc.) along the path that a user took in the user-determined timeframe prior to the creation of the immutable digital testimony as illustrated by immutable digital testimony (IDT) path identification operation **6205** in FIG. **6**. As explained below not all of the cameras have the ability to perform an IDT request process **2200** (FIG. **2**).

[0152] The IDT orchestrator **1210**, in some embodiments, gathers the information necessary to create a map of the IDT trail **6200** (e.g., using IDT geodata) as illustrated by create map of immutable digital testimony (IDT) trail operation **6210**. In addition, IDT orchestrator **1210** identifies and visually indicates the location of all third-party immutable digital testimonies (IDTs) and immutable digital testimony (IDT) cameras, as shown by identification operation **6220**. IDT orchestrator **1210**, in some embodiments, also performs a communication operation **69100** to send the visualization indicating the location of all third-party IDTs and IDT cameras to the device **1110** requesting the IDT trail.

[0153] The IDT orchestrator **1210**, in turn, performs a recording operation **6225** in conjunction with a communication operation **69200** to record on the IDT authentication blockchain **1330** the IDT ID, the IDT geodata related to the IDT trail, and all other immutable digital testimony identifiers identified or associated with other third-party IDTs and IDT cameras along the IDT trail.

[0154] In some embodiments, the IDT trail identifies other immutable digital testimonies (IDTs) created along a path. This method involves creating a map of the immutable digital testimony trail using IDT orchestrator **1210**. Generally, IDT geodata is used to identify third-party IDTs and cameras along a user's path during a predetermined time (e.g., 60 minutes) before the creation of an immutable digital testimony. The method includes an identification operation **6220** that performs identifying and visually indicating the locations of these third-party IDTs and cameras. Subsequently, communication operation **69100** sends the map of the immutable digital testimony trail to the device. The IDT orchestrator performs a record operation **6225** in conjunction with communication operation **69200** to record the IDT ID, geodata, and the IDs of the identified third-party IDTs and cameras on the IDT authentication blockchain **1330**.

[0155] Referring to FIG. **2**, FIG. **3** and FIG. **6**, in some embodiments, a thumbnail of the specific immutable digital testimony ID will appear in an immutable digital testimony (IDT) gallery for the testimony creator (e.g., on the witness layer **1100**) and, as applicable, in an IDT gallery for the testimony requestor who initiated a third-party immutable digital testimony (IDT) request process **2210**, as shown by presentation operation **3195**. In turn, in the event available, the IDT trail will be associated to a specific immutable digital testimony ID and be accessible from an IDT gallery.

[0156] All third-party IDTs and devices (e.g., mobile devices, cameras, etc.) that allow for the requesting of an immutable digital testimony view will be identified visually (e.g., by a distinguishable device icon, for example, by the color or size of the icon), as shown by device **6231**.

[0157] All third-party IDTs and cameras that do not provide the ability to request an immutable digital testimony view will be identified visually (e.g., by a distinguishable device icon, for example, by the color or size of the icon), as shown by device **6232**. In turn, the user can initiate an IDT request view process **2220** such or testimony share process **2230**, by selecting any of the devices (in this example cameras), as illustrated by third-party select operation **6110** of FIG. **6**. Next, upon selecting a device (e.g., camera) on the IDT trail, an IDT request process **6120** is initiated to initiate the IDT request.

[0158] Another aspect provides a process for generating an IDT trail. The method involves creating an immutable digital testimony trail that identifies immutable digital testimony created along a path, involving: a IDT path identification operation **6205** that performs creating, by an IDT orchestrator **1210**, a map of an immutable digital testimony trail by identifying (i) one or more third-party immutable digital testimonies, (ii) one or more cameras, or (iii) a combination of both (i) and (ii), each along a path taken by a user during a user-predetermined time prior to an immutable digital testimony creation. An identification operation **6220** performs identifying and visually indicating a location of (i) the one or more third-party immutable digital testimonies, (ii) the one or more cameras, or (iii) a combination of both (i) and (ii). A recording operation **6225** in conjunction with a communication operation **69200** perform recording on the IDT authentication blockchain **1330**, by the IDT orchestrator **1210**: the IDT trail associated with (i) an IDT ID and (ii) IDT geodata, each associated with the creator of the IDT local media, and one or more of an IDT IDs corresponding to (i) the one or more third-party immutable digital testimonies, (ii) the one or more cameras, or (iii) both (i) and (ii), correspondingly.

[0159] Referring to FIG. **3**, in some embodiments, the method further involves a presentation operation **3195** that performs presenting a thumbnail corresponding to a specific IDT ID on an IDT gallery for the creator of the immutable digital testimony, and a presentation operation **6100** performs presenting the IDT trail as part of metadata of the IDT local media in the IDT gallery, wherein any one or a combination of (i) the one or more third-party immutable digital testimonies and (ii) the one or more cameras, each of which allow for the requesting of an immutable digital testimony view are identified with one or more icons, correspondingly **6231**, **6232**. An IDT request initiation operation **6120** performs initiating an IDT viewing process **2220** or IDT sharing request process **2230** for a corresponding immutable digital testimony.

[0160] FIG. **4** illustrates a third-party request process **400** for requesting an immutable digital testimony, according to an example embodiment.

[0161] In some embodiments, a device (e.g., on the witness layer **1100**) executes instructions causing the device to perform an identification and authentication of a third-party immutable digital testimony requestor who has requested an immutable digital testimony via a third-party immutable digital testimony device, as illustrated by user authentication operation **4100**. In this example, the third-party IDT requestor is making the request via a device having installed therein IDT application **120**, the third-party device being referred to as third-party IDT device **1110-2**. The request for the immutable digital testimony causes an immutable digital testimony (IDT) request process to perform capturing the third-party immutable digital testimony requestor identifier (“third-party IDT ID”) and the device identifier (ID) of the third-party IDT device **1110-2** used to create the request for the immutable digital testimony.

[0162] Referring also to FIG. **2**, in turn, third-party immutable digital testimony requestor can initiate any one of a plurality of immutable digital testimony requests, as illustrated by initiate IDT request operation **4110**. In some embodiments, the request is to initiate a third-party IDT request process **2210**, requesting an IDT creator to initiate and share a new immutable digital testimony. In some embodiments, the request is to initiate an IDT viewing process **2220**, requesting an IDT creator permission to view one or more existing immutable digital testimonies of the IDT creator. In some embodiments, the request is to initiate an IDT sharing request process **2230**, requesting an IDT creator for permission to share one or more existing immutable digital testimonies of the IDT creator with another user or device on the network **100**.

[0163] In some embodiments, the third-party IDT device **1110-2**, creates the IDT request by combining the third-party IDT ID, the phone number of the IDT creator, and an IDT ID (e.g., in the case of an IDT viewing process **2220** and IDT sharing request process **2230**), as illustrated by create immutable digital testimony request operation **4120**.

[0164] In turn, the third-party IDT device **1110-2** sends the IDT request to the IDT orchestrator as illustrated by communication operation **49100**.

[0165] In an example embodiment, a method for requesting an immutable digital testimony is provided. The method involves identifying the user of a device, such as a third-party IDT device **1110-2**, to authenticate a third-party immutable digital testimony (IDT) requestor, by capturing a third-party immutable digital testimony requestor's immutable digital testimony creator ID, and a third-party immutable digital testimony requestor's immutable digital testimony device ID of the third-party device used to create the immutable digital testimony request, as illustrated by user authentication operation **4100**. The method further involves initiating, by the third-party immutable digital testimony requestor from an immutable digital testimony creator, any one or more of: an immutable digital testimony request requesting to initiate and share a new immutable digital testimony, an immutable digital testimony view request requesting permission to view an immutable digital testimony creator's existing immutable digital testimony, an immutable digital testimony share request requesting permission to share one of immutable digital testimony creator's existing immutable digital testimonies with another immutable digital testimony network user, as illustrated by initiate IDT request operation **4110**. The method further involves creating, by the device (e.g., the third-party IDT device **1110-2** in FIG. **4**), the immutable digital testimony request by combining any one or a combination of: the third-party immutable digital testimony requestor's ID, the immutable digital testimony creator's phone number, and immutable digital testimony ID as illustrated by create immutable digital testimony request operation **4120**. In turn, the method performs sending, by the third-party IDT device **1110-2**, the immutable digital testimony request to an immutable digital testimony orchestrator as illustrated by communication operation **49100**.

[0166] In some embodiments, authenticating the third-party IDT requestor includes capturing a third-party IDT requestor ID, and a third-party IDT requestor device ID corresponding to a device of the third-party IDT requestor used to create the immutable digital testimony request, as illustrated by user authentication operation **4100**.

[0167] In some use cases, the IDT orchestrator **1210** (e.g., on orchestrator layer **1200**) directs the third-party IDT request to a user that is not an IDT creator. If the user is not a member of the network **100**, then the IDT orchestrator **1210** prompts the user (referred to as “an unregistered IDT creator”) to download the IDT application **120** and register to create one or more IDTs, as shown by communication operation **49200**.

[0168] In some embodiments the device of the IDT creator or intended IDT creator performs an incoming IDT request operation **4130** to present an alert for the IDT creator (or intended IDT creator) of an incoming immutable digital testimony (IDT) request. Here the device is referred to as IDT creator device **1110-1** or simply creator device **1110-1**, which also has installed thereon IDT application **120**.

[0169] In turn, the IDT creator (or intended IDT creator) evaluates whether to grant the IDT request as illustrate by evaluation of IDT request operation **4140**.

[0170] If the IDT creator (or intended IDT creator) declines the IDT request, then the device notifies the denial as illustrated by communication operation **41410** to the IDT orchestrator **1210**, which in turn notifies the requestor, as illustrated by communication operation **41420**.

[0171] If the IDT creator (or intended IDT creator) approves the request for an IDT as illustrated by approval of the third-party IDT request operation **4150**, then the IDT creator initiates the IDT creation process as illustrated by initiated IDT creation operation **4151**.

[0172] If the third-party request is a request for approval to view an IDT and the IDT creator approves the request, as illustrated by approval of IDT view operation **4160**, then the creator device **1110-1** notifies the IDT orchestrator **1210** of the approval as illustrated via communication operation **41610** and the IDT orchestrator **1210** updates the permission management system **3210** on the orchestrator layer **1200** as illustrated by update operation **4210**, granting the third-party requestor, based on the third-party requestor ID, viewing rights over the specific IDT ID that was included in the IDT view request.

[0173] In turn, the creator device **1110-1** records the view permission as illustrated by communication to the IDT authentication blockchain **1330** on the IDT layer **1300**, creating a log entry indicating that the third-party requestor ID has viewing rights over the specific IDT ID that was included in the IDT view request as illustrated by communication operation **49300**.

[0174] In the event the IDT creator approves a request to share an immutable digital testimony (e.g., IDT sharing request **2300**), as illustrated by operation **4170**, then the creator device **1110-1** notifies the IDT orchestrator **1210** of the approval as illustrated by communication operation **41710** and IDT orchestrator **1210** performs an update operation **4220** to update the permission management system **3210** on the orchestrator layer **1200**, granting the third-party requestor, via the third-party requestor ID associated with the third-party request, sharing rights over the specific IDT ID that was included in the IDT share request.

[0175] In turn, the creator device **1110-1** records the share permission to the IDT authentication blockchain **1330** on the IDT layer **1300**, creating a log entry indicating that the third-party requestor associated with the third-party requestor ID has sharing rights over the specific IDT ID that was included in the IDT share request as illustrated by communication operation **49400**.

[0176] In some embodiments, a thumbnail of the specific IDT ID (that was included in the IDT request) is presented via an immutable digital testimony (IDT) gallery for the IDT creator (e.g., on witness layer **1100**) as illustrated by presentation operation **4180** and present operation **4190** presents the thumbnail in the IDT gallery for the third-party requestor who initiated the IDT request. In an example implementation, the thumbnail will have a badge of a first color (e.g., green) if viewing permission was granted and a badge of another color (e.g., yellow) if sharing permission was granted. Other indication techniques can be used instead and still be within the scope of the embodiments herein.

[0177] FIG. **5**. illustrates an immutable digital viewing process **500**, according to an example embodiment.

[0178] In an example implementation, a user opens, via a device (e.g., on witness layer **1100**) such as creator device **1110-1** or third-party IDT device **1110-2**, an IDT gallery feature as illustrated by open IDT gallery operation **5100**. To view an IDT, the user selects the specific thumbnail. The device receives a selection of a specific thumbnail as illustrated by user IDT selection operation **5110**.

[0179] In turn, the device sends the request to view an immutable digital testimony (for a specific IDT ID) to the IDT orchestrator **1210** (on orchestrator layer **1200**), as illustrated by communication operation **59100**.

[0180] The IDT orchestrator **1210** receives the request to view the IDT and verifies on the IDT authentication blockchain **1330** that the user associated with the request for the IDT view has view permission as illustrated by communication, as illustrated by communication operation **59200**. If the user associated with the request has viewing permission, the smart contract on the IDT authentication blockchain **1330** retrieves the IDT frame hashing scheme from the IDT metadata recorded on the IDT authentication blockchain **1330** as illustrated by communication operation **59250**. If the user has viewing permission, the smart contract on the IDT authentication blockchain **1330** notifies the IDT orchestrator **1210** and sends the IDT cloud address, the live stream hash, and IDT frame hashing scheme, to the device via IDT orchestrator **1210**, as illustrated by communication operation **59300** and communication operation **59400**, respectively.

[0181] If the IDT authentication blockchain **1330** also has an alternate immutable digital testimony, then the IDT orchestrator **1210** will also send to the device the corresponding information for both the corresponding IDT ID and the immutable digital testimony alternate ID.

[0182] If the user is not authorized to view the immutable digital testimony, then the IDT authentication blockchain **1330** notifies, via communication operation **59300**, IDT orchestrator **1210**. In turn, IDT orchestrator notifies the IDT application **120** on the device that there is no viewing permission via communication operation **59500** and issues an inquiry as to whether the user wants to send a request to view an immutable digital testimony (e.g., by initiating an IDT viewing process **2220**).

[0183] A viewing authorization operation **5115**, in turn, performs a viewing authorization validation. The device (e.g., on the witness layer **1100**) then performs a request media operation **5120** to request the IDT cloud media from the IDT cloud **1310**.

[0184] IDT cloud **1310**, in turn, transmits (e.g., streams or downloads) the IDT cloud media directly from the IDT cloud address on the IDT cloud **1310**, as illustrated by communication operation **59600**.

[0185] As the media is transmitted, a media validation operation **5130** performs validating the incoming IDT cloud media by creating a viewing hash. A "viewing hash" as used herein generally means a hash of media that is played back live to a viewer to validate that the original media has not been corrupted during transmission nor been altered since the date of its corresponding immutable digital testimony creation. In an example implementation, a viewing has is created by hashing each frame of the streamed IDT cloud media (using the IDT frame hashing scheme, e.g., from communication operation **59400** which are then combined or hashed together to form the viewing hash. Once the viewing hash is completed it is compared with the live stream hash.

[0186] In some embodiments, if the viewing hash matches the live stream hash, then an IDT display operation **5140** performs presenting the IDT media on the IDT viewer and the device records the viewing event on the IDT authentication blockchain **1330** including the IDT ID the IDT user ID, the IDT device ID, and date of viewing via communication operation **59700**.

[0187] In some embodiments, if there is an alternate IDT then the media corresponding to the alternate IDT will also be transmitted from the alternate IDT DFS address and the viewing hash is created by applying the hashing procedure to the alternate IDT media and comparing it to the local symbiotic hash (which would be downloaded in place of the live stream hash).

[0188] If the viewing hash does not match the live stream hash, then the user is alerted, via the device, that the streaming IDT cloud media has been altered or the streaming IDT cloud media was corrupted during the download, requiring reinitiating the transmission as illustrated by reinitiating operation **5150**.

[0189] The IDT DFS Address (and the corresponding media) can potentially be used to recreate or reconstruct an IDT in case the IDT Cloud Media is corrupted or altered.

[0190] FIG. **7** illustrates an immutable digital testimony (IDT) authentication process **700**, according to an example embodiment. Generally, IDT authentication process **700** enables a user on the network **100** to request that an immutable digital testimony be sent and authenticated for the benefit of a specific user.

[0191] The method involves an initiation operation **7110** that performs receiving an initiation request (e.g., via user input device **106** of device **1110**), the initiation request requesting that the immutable digital testimony of the user issuing the request, such as the IDT creator, be sent and authenticated for the benefit of a specific user. In turn, the user selects the thumbnail from their photo gallery and select the authenticate button as illustrated by select operation **7111**.

[0192] A recipient selection operation **7112** receives (e.g., via user input device **106** of device **1110**) a selection of a person or persons (e.g., from a contacts list on the device) to whom the user wants to send the immutable digital testimony authentication. The recipient is referred to as an "intended recipient."

[0193] A create immutable digital testimony operation **7120** performs creating, by the device, the IDT authentication and sending, in an example implementation, the intended recipient's phone number and IDT ID to the IDT orchestrator **1210** as illustrated by communication operation **79100**.

[0194] Upon receipt of the intended recipient's phone number and IDT ID, IDT orchestrator **1210** proceeds to execute the Immutable digital testimony authentication. In an example implementation, if the intended recipient is not a member of the network **100** (e.g., the intended recipient has not registered for access to the network **100**), then the IDT orchestrator **1210** will prompt the intended recipient to download and execute on their device the IDT application **120**, as illustrated by communication operation **79200**.
[0195] IDT orchestrator **1210**, in turn, performs a recording operation to record the IDT authentication on the IDT authentication blockchain **1330**, as illustrated by communication operation **79300**. Recording operation further creates a log entry indicating that the intended recipient's IDT user ID has viewing rights over a specific IDT ID that was shared.
[0196] The IDT orchestrator **1210**, in turn, retrieves from the IDT authentication blockchain **1330**, the IDT cloud address, the live stream hash, and the IDT frame hashing scheme, as illustrated by communication operation **79400**.
[0197] If the IDT authentication blockchain **1330** also has an alternative immutable digital testimony, then the IDT orchestrator **1210** will also send to the intended recipient the corresponding information for both the IDT ID and the IDT alternate ID.
[0198] IDT orchestrator **1210**, in some embodiments, further performs an update operation **7230** to update the permission management system on the orchestrator layer **1200**, granting the intended Recipient's Immutable digital testimony User ID viewing rights over the specific IDT ID that was included in the IDT authentication.
[0199] The IDT orchestrator **1210** further sends, via communication operation **79500**, the IDT ID, IDT cloud address, the live stream hash, and the IDT frame hashing scheme to the device of the intended recipient.
[0200] In turn, the device of the intended recipient alerts the intended recipient of an incoming IDT authentication, as illustrated by incoming authentication alert operation **7130**. A presentation operation **7140** performs presenting a thumbnail of the specific IDT ID (e.g., that was included in the IDT authentication), causing the thumbnail to appear on the IDT gallery of the intended recipient.
[0201] The IDT application **120**, in turn, will (e.g., automatically) perform a request operation **7150** to request the IDT cloud media from the IDT cloud and the media will be transmitted (e.g., streamed or downloaded), as illustrated by communication operation **79600**.
[0202] As the media is being transmitted (e.g., streamed or downloaded), the incoming IDT cloud media is validated by a validation operation **7160**, by forming a validation hash. A “validation hash” as used herein generally means a hash of media that is played back live to a viewer to validate that the original media has not been altered since the date of its corresponding immutable digital testimony creation.
[0203] The validation hash is formed, in an example implementation, by hashing of each frame of the live streamed IDT cloud media (using the IDT frame hashing scheme received via communication operation **79500**) which are then combined or hashed together to form the validation hash. Once the validation hash is completed it is compared to the live stream hash.
[0204] If the validation hash matches the live stream hash, then a presentation operation **7170** performs displaying the immutable digital testimony on an IDT viewer and the device performs a recording operation **7171** in conjunction with communication operation **79700** to record the viewing event on the IDT authentication blockchain **1330** including the IDT ID, the IDT user ID, the IDT device ID, and date of viewing.
[0205] If an alternative immutable digital testimony exists, then the media corresponding to the alternative immutable digital testimony will also be transmitted (e.g., streamed or downloaded) from the alternate IDT cloud address and the validation hash is created by applying the hashing procedure to the alternate IDT media and comparing it to the local symbiotic hash (which, in an example implementation, would be downloaded in place of the live stream hash).

[0206] If the validation hash does not match the live stream hash, then the IDT application **120** alerts the intended recipient that the IDT cloud media has been altered or the IDT cloud media was corrupted during the transmission as illustrated by communication operation **71720**, requiring reinitiating the transmission (e.g., streaming or download), as illustrated by start operation **7180**.
[0207] The IDT DFS address (and the corresponding media) can, in some embodiments, also be used to recreate or reconstruct an immutable digital testimony in case the IDT cloud media is corrupted or altered.
[0208] In some embodiments, a non-transitory computer-readable medium having stored thereon one or more sequences of instructions for causing one or more processors to perform any of the methods described herein.
[0209] The following are additional clauses relative to the present disclosure, which could be combined and/or otherwise integrated with any of the embodiments described above or listed in the claims below. [0210] Clause 1. A method for processing a request for immutable digital testimony media, comprising: [0211] transmitting immutable digital testimony cloud media from one or more cloud addresses on a cloud (**59600**); [0212] as the immutable digital testimony cloud media is being transmitted, validating the immutable digital testimony cloud media (**5130**) by: [0213] selecting one or more frames of the immutable digital testimony cloud media that is being transmitted, [0214] hashing one or more frames of the selected one or more frames of the transmitting immutable digital testimony cloud media using a selected immutable digital testimony hashing scheme, thereby creating one or more immutable digital testimony transmitting hashes, and [0215] combining the one or more immutable digital testimony transmitting hashes, thereby forming a viewing hash; and [0216] comparing the viewing hash with a live stream hash, wherein: [0217] (a) if the viewing hash matches the live stream hash: playing the immutable digital testimony cloud media on an immutable digital testimony viewer (**5140**) and recording, by the device, a viewing event on an immutable digital testimony authentication blockchain including an immutable digital testimony ID, third-party requester immutable digital testimony ID, a third-party requester immutable digital testimony device ID, and a date of viewing (**59700**), and [0218] (b) if the viewing hash does not match the live stream hash: generating an alert indicating that the immutable digital testimony cloud media has been altered or the immutable digital testimony cloud media was corrupted during the transmitting, thereby requiring reinitiating a download of the immutable digital testimony cloud media (**5150**). [0219] Clause 2. The method according to clause 1, wherein the live stream hash (**3130**) was created by: [0220] hashing one or more frames of the immutable digital testimony stream media for the one or more cameras by applying a selected immutable digital testimony hashing scheme from a plurality of hashing schemes, thereby creating one or more immutable digital testimony stream hashes (**3130**, **3132**); and [0221] forming the live stream hash (**3133**) from the one or more immutable digital testimony stream hashes. [0222] Clause 3. The method according to claim clause 1, further comprising: [0223] transmitting (**59600**) immutable digital testimony cloud media of an alternate immutable digital testimony from one or more immutable digital testimony cloud addresses or one or more immutable digital testimony DFS addresses; and [0224] displaying the immutable digital testimony cloud media corresponding to the alternate immutable digital testimony. [0225] Clause 4. A method for validating media stored on a cloud server, comprising: [0226] selecting one or more frames of a live stream and hashing one or more frames of the selected one or more frames of the live stream using a selected hashing scheme, thereby creating one or more live stream hashes; [0227] combining the one or more live stream hashes, thereby forming a combined live stream media hash; [0228] sending the combined live stream media hash to a smart contract; [0229] selecting one or more frames of media stored on a first storage device, and hashing one or more frames of the selected frames of the media stored on the first storage device using the selected immutable digital testimony hashing scheme, thereby creating one or more first storage device media hashes, and [0230] combining the one or more first storage device media hashes, thereby forming a combined first storage device media hash; [0231]

selecting one or more frames of media stored on a second storage device, and hashing one or more frames of the selected frames of the media stored on the second storage device using the selected immutable digital testimony hashing scheme, thereby creating one or more second storage device media hashes, and [0232] combining the one or more second storage device media hashes, thereby forming a combined second storage device media hash; [0233] comparing, by the smart contract: the combined live stream media hash, the combined first storage device media hash, and the combined second storage device media hash; [0234] if a) the combined live stream media hash mirrors the combined first storage device media hash and b) the combined live stream media hash mirrors the combined second storage device media hash: [0235] providing a notification, by the smart contract, that the live stream media, the media stored on the first storage device and the media stored on the second storage device has been validated; and [0236] if a) the combined live stream media hash and the combined first storage device media hash do not match or b) the combined live stream media hash and the combined second storage device media hash do not match: [0237] providing a notification, by the smart contract, that one or a combination of (i) the live stream media, (ii) the media stored on the first storage device, and (iii) the media stored on the second storage device, has been corrupted. [0238] Clause 5. A system having one or more processors communicatively coupled to a plurality of sensors, and a non-transitory memory, the non-transitory memory storing instructions that when executed by the processor, control the system to perform the methods of Clauses 1-4. [0239] Clause 6. A non-transitory computer-readable medium having stored thereon one or more sequences of instructions for causing one or more processors to perform the methods of Clauses 1-4.
[0240] While various example embodiments of the present invention have been described above, it should be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant art(s) that various changes in form and detail can be made therein. Thus, the present invention should not be limited by any of the above described example embodiments, but should be defined only in accordance with the following claims and their equivalents.
[0241] In addition, it should be understood that the FIGS. **1-7** are presented for example purposes only. The architecture of the example embodiments presented herein is sufficiently flexible and configurable, such that it may be utilized (and navigated) in ways other than that shown in the accompanying figures.
[0242] Further, the purpose of the foregoing Abstract is to enable the U.S. Patent and Trademark Office and the public generally, and especially the scientists, engineers and practitioners in the art who are not familiar with patent or legal terms or phraseology, to determine quickly from a cursory inspection the nature and essence of the technical disclosure of the application. The Abstract is not intended to be limiting as to the scope of the example embodiments presented herein in any way. It is also to be understood that the procedures recited in the claims need not be performed in the order presented.

Claims

1. A method for creating and validating an immutable digital testimony: activating one or more cameras associated with a device having a device identifier; capturing media through the one or more cameras of the device to generate stream media; during generation of the stream media: hashing one or more frames of the stream media using a hashing scheme to create a plurality of stream hashes; forming, by the device, a live stream hash from the plurality of stream hashes; transmitting the stream media to at least one of: a cloud server to form cloud media, or local storage on the device to form local media; validating the stream media by: forming at least one of: a cloud media hash from the cloud media using the hashing scheme, or a local symbiotic hash from the local media using the hashing scheme; and comparing the live stream hash against at least one of: the cloud media hash, or the local symbiotic hash; and upon confirming the comparison of the live stream hash matches the cloud media hash or the local symbiotic hash, correspondingly: creating an immutable digital testimony hash, and recording the immutable digital testimony hash on a blockchain.

2. The method of claim 1, further comprising: encrypting the stream media to create encrypted media; transmitting the encrypted media to a distributed file system; receiving distributed file system addresses indicating storage locations of the encrypted media; and recording the distributed file system addresses on the blockchain.

3. The method of claim 1, further comprising: sending, by the device to a smart contract, the live stream hash for validation; creating, by the smart contract, an immutable digital testimony identifier; and recording the immutable digital testimony identifier on the blockchain.

4. The method of claim 1, wherein upon confirming either: the live stream hash matches the cloud media hash, or the live stream hash matches the local symbiotic hash, using a smart contract to: record a plurality of user permissions on an authentication blockchain; send the user permissions to an orchestrator to create a log entry indicating access rights; grant viewing permission for the immutable digital testimony to any one of (i) a creator, (ii) a requestor, or (iii) both the creator and the requestor; and periodically validate authenticity using the immutable digital testimony hash recorded on a public blockchain.

5. The method of claim 1, wherein upon determining either: the live stream hash and cloud media hash do not match, or the live stream hash and local symbiotic hash do not match, using a smart contract to: create an alternate immutable digital testimony identifier; create an alternate immutable digital testimony hash using the local media; record the alternate immutable digital testimony hash on the authentication blockchain; create an alternate immutable digital testimony based on the local media; record a plurality of alternate addresses for the alternate immutable digital testimony; and provide notification that the stream media was corrupted during transmission.

6. The method of claim 1, further comprising: creating an immutable digital testimony trail associated with the immutable digital testimony by: identifying, during a user-determined timeframe prior to creating the immutable digital testimony, third-party immutable digital testimonies and cameras along a path taken by a user of the device; creating, by an immutable digital testimony orchestrator, a map displaying: the path taken by the user, locations of the identified third-party immutable digital testimonies, and locations of the identified cameras; sending the map to the device for display; recording on the blockchain: an association between the immutable digital testimony trail and the immutable digital testimony, identifiers of the identified third-party immutable digital testimonies and cameras, and geodata indicating the path taken by the user.

7. The method of claim 1, wherein validating the stream media further comprises: comparing the live stream hash to both the cloud media hash and the local symbiotic hash; and upon confirming that the live stream hash matches both the cloud media hash and the local symbiotic hash, creating and recording the immutable digital testimony hash on the blockchain.

8. A system for creating and validating an immutable digital testimony, comprising: one or more processors; one or more cameras associated with a device having a device identifier; a memory storing instructions that, when executed by the one or more processors, cause the system to: activate the one or more cameras; capture media through the one or more cameras to generate stream media; during generation of the stream media: hash one or more frames of the stream media using a hashing scheme to create a plurality of stream hashes; form a live stream hash from the plurality of stream hashes; transmit the stream media to at least one of: a cloud server to form cloud media, or local storage on the device to form local media; validate the stream media by: forming at least one of: a cloud media hash from the cloud media scheme using the hashing scheme, or a local symbiotic hash from the local media scheme using the hashing scheme; and comparing the live stream hash against at least one of: the cloud media hash, or the local symbiotic hash; and upon

confirming the comparison of the live stream hash matches the cloud media hash or the local symbiotic hash, correspondingly: create an immutable digital testimony hash, and record the immutable digital testimony hash on a blockchain.
9. The system of claim 8, wherein the instructions further cause the system to: encrypt the stream media to create encrypted media; transmit the encrypted media to a distributed file system; receive distributed file system addresses indicating storage locations of the encrypted media; and record the distributed file system addresses on the blockchain.
10. The system of claim 8, wherein the instructions further cause the system to: send the live stream hash to a smart contract for validation; create, by the smart contract, an immutable digital testimony identifier; and record the immutable digital testimony identifier on the blockchain.
11. The system of claim 8, wherein upon confirming either: the live stream hash matches the cloud media hash, or the live stream hash matches the local symbiotic hash, the instructions further cause the system to use a smart contract to: record a plurality of user permissions on an authentication blockchain; send the user permissions to an orchestrator to create a log entry indicating access rights; grant viewing permission for the immutable digital testimony to any one of (i) a creator, (ii) a requestor, or (iii) both the creator and the requestor; and periodically validate authenticity using the immutable digital testimony hash recorded on a public blockchain.
12. The system of claim 8, wherein upon determining either: the live stream hash and cloud media hash do not match, or the live stream hash and local symbiotic hash do not match, the instructions further cause the system to use a smart contract to: create an alternate immutable digital testimony identifier; create an alternate immutable digital testimony hash using the local media; record the alternate immutable digital testimony hash on the authentication blockchain; create an alternate immutable digital testimony based on the local media; record a plurality of alternate addresses for the alternate immutable digital testimony; and provide notification that the stream media was corrupted during transmission.
13. The system of claim 8, wherein the instructions further cause the system to: create an immutable digital testimony trail associated with the immutable digital testimony by: identifying, during a user-determined timeframe prior to creating the immutable digital testimony, third-party immutable digital testimonies and cameras along a path taken by a user of the device; creating, by an immutable digital testimony orchestrator, a map displaying: the path taken by the user, locations of the identified third-party immutable digital testimonies, and locations of the identified cameras; sending the map to the device for display; recording on the blockchain: an association between the immutable digital testimony trail and the immutable digital testimony, identifiers of the identified third-party immutable digital testimonies and cameras, and geodata indicating the path taken by the user.
14. The system of claim 8, wherein validating the stream media further comprises: comparing the live stream hash to both the cloud media hash and the local symbiotic hash; and upon confirming that the live stream hash matches both the cloud media hash and the local symbiotic hash, creating and recording the immutable digital testimony hash on the blockchain.
15. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to: activate one or more cameras associated with a device having a device identifier; capture media through the one or more cameras to generate stream media; during generation of the stream media: hash one or more frames of the stream media using a hashing scheme to create a plurality of stream hashes; form a live stream hash from the plurality of stream hashes; transmit the stream media to at least one of: a cloud server to form cloud media, or local storage on the device to form local media; validate the stream media by: forming at least one of: a cloud media hash from the cloud media using the hashing scheme, or a local symbiotic hash from the local media using the hashing scheme; and comparing the live stream hash against at least one of: the cloud media hash, or the local symbiotic hash; and upon confirming the comparison of the live stream hash matches the cloud media hash or the local symbiotic hash,

correspondingly: create an immutable digital testimony hash, and record the immutable digital testimony hash on a blockchain.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the one or more processors to: encrypt the stream media to create encrypted media; transmit the encrypted media to a distributed file system; receive distributed file system addresses indicating storage locations of the encrypted media; and record the distributed file system addresses on the blockchain.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the one or more processors to: send the live stream hash to a smart contract for validation; create, by the smart contract, an immutable digital testimony identifier; and record the immutable digital testimony identifier on the blockchain.

18. The non-transitory computer-readable medium of claim 15, wherein upon confirming either: the live stream hash matches the cloud media hash, or the live stream hash matches the local symbiotic hash, the instructions further cause the one or more processors to use a smart contract to: record a plurality of user permissions on an authentication blockchain; send the user permissions to an orchestrator to create a log entry indicating access rights; grant viewing permission for the immutable digital testimony to any one of (i) a creator, (ii) a requestor, or (iii) both the creator and the requestor; and periodically validate authenticity using the immutable digital testimony hash recorded on a public blockchain.

19. The non-transitory computer-readable medium of claim 15, wherein upon determining either: the live stream hash and cloud media hash do not match, or the live stream hash and local symbiotic hash do not match, the instructions further cause the one or more processors to use a smart contract to: create an alternate immutable digital testimony identifier; create an alternate immutable digital testimony hash using the local media; record the alternate immutable digital testimony hash on the authentication blockchain; create an alternate immutable digital testimony based on the local media; record a plurality of alternate addresses for the alternate immutable digital testimony; and provide notification that the stream media was corrupted during transmission.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the one or more processors to: create an immutable digital testimony trail associated with the immutable digital testimony by: identifying, during a user-determined timeframe prior to creating the immutable digital testimony, third-party immutable digital testimonies and cameras along a path taken by a user of the device; creating, by an immutable digital testimony orchestrator, a map displaying: the path taken by the user, locations of the identified third-party immutable digital testimonies, and locations of the identified cameras; sending the map to the device for display; recording on the blockchain: an association between the immutable digital testimony trail and the immutable digital testimony, identifiers of the identified third-party immutable digital testimonies and cameras, and geodata indicating the path taken by the user.

21. The non-transitory computer-readable medium of claim 15, wherein validating the stream media further comprises: comparing the live stream hash to both the cloud media hash and the local symbiotic hash; and upon confirming that the live stream hash matches both the cloud media hash and the local symbiotic hash, creating and recording the immutable digital testimony hash on the blockchain.